

EngageIQ — Smart Engagement Opportunity Scorer

BAX-423 Big Data · Spring 2026 · Yang, Alice · UC Davis GSM

Problem Statement

Developers and founders waste hours manually scanning GitHub and Hacker News to find where to contribute, comment, or build reputation. EngageIQ automates this by ingesting 10,046 opportunities across 15 technical domains, scoring them via semantic embeddings and community signals, and adapting rankings in real time from user feedback.

System Architecture

Stage	Component	Lecture
1 · Ingestion	GitHub API + Algolia/HN API; Bloom Filter dedup; Python queue refresh Real HN stories via domain keyword queries (hn_item URLs, 0 fallbacks)	Lec 2
2 · Embedding	Sentence-BERT all-MiniLM-L6-v2 (384-dim); FAISS IndexFlatIP <2 ms	Lec 5
3 · Intent	7-intent classifier from role + interests; adaptive query expansion + candidate injection	app.py
4 · Scoring	Composite: 0.40 relevance + 0.30 community + 0.20 visibility + 0.10 (1-effort)	Lec 7
5 · Ranking	Multi-stage: intent-aware rerank + diversity cap + Thompson Sampling bandit	Lec 7/8
6 · Adaptation	Thompson Sampling Beta-Bernoulli bandit; domain preference learning over 50+ rounds	Lec 8
7 · Analytics	Pandas batch: domain health, trending repos, volume-over-time	Lec 3
8 · Dashboard	Streamlit: ranked cards, Why this?, suggested actions, CSV/JSON export	—

Dataset

Metric	Value
Total records	10,046
Technical domains	15
GitHub records	8,588 — real API-derived issues + repos; direct github.com links
HN records	1,458 — real Algolia/HN API stories; direct news.ycombinator.com/item?id= URLs; 0 fallbacks
Storage	CSV offline snapshot (committed to repo) + SQLite for live records
Live / offline split	10,046 offline snapshot; live records added via on-demand refresh

BAX-423 Techniques & Benchmarks

Integrated BAX-423 Techniques

Technique	Lec	File	Role	Benchmark
Bloom Filter (sketching/dedup)	2	bloom_filter.py	Rejects duplicate streamed records before DB insert	Drop rate on 10,046 unique IDs
Sentence-BERT + FAISS IndexFlatIP	5	embeddings.py	384-dim semantic embeddings; exact inner-product search	<2 ms at 10k scale NDCG@10 emb-only: 0.49
Multi-stage Ranking (NDCG@10)	7	ranking.py	Composite score + diversity rerank; beats stars-only	NDCG@10 0.2425 vs stars-only: 0.1569
Thompson Sampling (RL bandit)	8	adaptive_learning.py	Beta-Bernoulli posterior per opportunity; domain-specific MLs from feedback	50 rounds, MLs from feedback

Ranking Benchmark — NDCG@10 (Sofia Persona, n=500, seed=42)

Quality-aligned graded relevance: in-domain records scored 0-1 by stars (40%) + activity (30%) + community health (20%) + GFI bonus (20%). Higher NDCG@10 = better (max = 1.0). Random stratified sample, seed=42.

Ranking Method	NDCG@10	Notes
Embedding Similarity Only	0.4915	Strong domain retrieval for aligned queries
Full Composite + Re-rank *	0.2425	Composite scoring + diversity; beats stars-only by 55%
Stars-Only Ranking	0.1569	Popularity != persona relevance
Random Baseline	0.1036	No signal — lower bound

* Benchmark isolates composite scoring from intent-aware reranking. In the live app, Full Composite also applies intent-specific GFI/diversity reranking — the two stages are complementary.

Adaptive Learning — Thompson Sampling (50 Rounds)

Synthetic Sofia persona: engage if domain in {ML, AI Research}, skip otherwise. Thompson Sampling Beta posterior updated each round. Random seed = 42.

Metric	Before (Round 0)	After (Round 50)
ML + AI Research domain pref score	0.50 (uniform prior)	1.00 (maximum)
Other 13 domains avg pref score	0.50 (uniform prior)	0.00 (minimized)
Preferred / non-preferred score gap	0.00	+1.00

6 Core Capabilities & Persona Test Results

6 Core Capabilities

#	Capability	Implementation
1	Multi-Source Ingestion & On-Demand Refresh	GitHub API + Hacker News API; Bloom Filter dedup; Python queue-based refresh
2	Content Embedding & Similarity Retrieval	all-MiniLM-L6-v2 (384-dim); FAISS IndexFlatIP; embedding cache for <2 ms queries
3	Engagement Scoring & Multi-Stage Ranking	4-component composite + diversity rerank; NDCG@10 = 0.2425 vs 0.1036 random baseline
4	Adaptive Learning from Feedback	Thompson Sampling Beta-Bernoulli bandit; domain pref 0.50->1.00 over 50 rounds
5	Batch Analytics & Trend Detection	Pandas batch: domain health, trending repos, volume-over-time, rising opportunities
6	Dashboard & Engagement Brief	Streamlit: ranked cards, Why this?, suggested actions; CSV/JSON export

Persona Pass/Fail Test Results

Each persona tested against all 6 capabilities. Pass = feature surfaces expected results for that persona's interests.

Persona / Role	1 Ingest	2 Embed	3 Rank	4 Adapt	5 Analytics	6 Brief	Result
Sofia — ML Student / Portfolio	Pass	Pass	Pass	Pass	Pass	Pass	PASS
David — DevOps / Niche Comm.	Pass	Pass	Pass	Pass	Pass	Pass	PASS
Lina — Data Journalist / Trends	Pass	Pass	Pass	Pass	Pass	Pass	PASS
Raj — Startup Founder / B2B	Pass	Pass	Pass	Pass	Pass	Pass	PASS

Evidence: Sofia top-10 surfaces ML/AI Research repos (gfi>=6, domain_match=8). David leads with DevOps/K8s and Cloud API repos (infra_kw=4, domain_match=9). Lina includes trending open-source sorted by growth_rate (avg_trend=0.69). Raj highlights Developer Tools and B2B SaaS by community engagement (api=6, dm=10).

Hidden Persona Robustness — Adaptive Intent Layer (11/11 PASS)

The same intent inference, adaptive query expansion, and intent-aware reranking layer powers both the main Action Queue and the Persona Test Panel. Stress-tested against 11 roles not in the system's ROLE_INTENT map — intent inferred from role + interest keywords; evaluated against multi-condition pass criteria (domain_match>=4, primary_match>=1, src_fit>=6, neg=0).

Hidden Persona	Inferred Intent	domain	primary	src_fit	neg	Result
Security Researcher	security_review	5/10	5/10	10/10	0	PASS
Climate Tech Founder	startup_growth	7/10	4/10	10/10	0	PASS
Beginner Developer	contribution	5/10	4/10	10/10	0	PASS
Open Source Maintainer	community_engagement	8/10	4/10	10/10	0	PASS
Product Manager	startup_growth	7/10	7/10	10/10	0	PASS
Mobile Developer	mobile_contribution	8/10	8/10	10/10	0	PASS
Game Developer	generic	10/10	6/10	10/10	0	PASS
Data Engineer	data_engineering	6/10	5/10	9/10	0	PASS
Academic ML Researcher	generic	10/10	8/10	10/10	0	PASS

Education Creator	trend_spotting	10/10	9/10	10/10	0	PASS
Privacy Researcher	security_review	5/10	5/10	10/10	0	PASS

Limitations, Future Work & Submission Checklist

Honest Scope — Course-Project Implementation

Area	Current Implementation	Production Equivalent
Streaming	On-demand API refresh + Python queue/thread simulation	Apache Kafka / Flink persistent streaming cluster
Data Sources	GitHub + Hacker News (two sources meets spec; Reddit excluded — OAuth2 unavailable in scope)	Expand to Reddit PRAW, LinkedIn, DEV.to
AI Suggestions	Deterministic template-based engagement action generation (avoids API cost and hallucination risk)	LLM-generated actions via Claude / GPT with prompt caching
Batch Analytics	Pandas batch over 10,046-record offline snapshot (<1 s query latency at this scale)	Apache Spark / Dask for distributed processing at scale
GH Archive	build_real_dataset.py includes GH Archive support; not used in runtime pipeline	Scheduled GH Archive pulls as supplemental source

Future Work

- Connect suggested engagement actions to a live LLM (e.g., Claude Haiku) for natural-language drafts; cache generated briefs to minimize API cost.
- Expand data sources to Reddit (authenticated PRAW), LinkedIn, and DEV.to for broader community coverage.
- Add scheduled GH Archive pulls for historical trend analysis beyond real-time API limits.
- Extend Thompson Sampling to opportunity-type preferences (issue vs. repo vs. post) for finer-grained adaptive ranking.
- Replace Pandas batch analytics with Dask or Spark when dataset grows beyond 100k records.

Deployment Notes

App requires sentence-transformers and faiss-cpu (~450 MB combined). On first launch embeddings are computed once and cached to data/embeddings.npy (~16 MB). Subsequent launches load the cache in <1 second. Live API fetch is optional — the app runs fully on the pre-seeded offline dataset.

Submission Checklist

Item	Status
code/ — all .py files + requirements.txt	Included
data/opportunities.csv — 10,046 offline records	Included
data/embeddings.npy — pre-computed 384-dim embeddings	Included
brief.pdf — this document	Included
prompts.md — development + planned runtime prompts	Included
Live public URL	https://engageiq-bax423git-qianyingyang.streamlit.app
GitHub Repo	https://github.com/qyayang/engageiq-bax423
ZIP: Yang_Alice_BAX423_Final.zip	Per one-pager spec